

Validity Analysis: FOLIO

Target context: US Defense and High-Reliability Systems Engineers

Overall risk: **HIGH**

Deployment Context

We are a US engineering firm building a tool that identifies logical inconsistencies across technical specification documents for American systems engineers. The tool analyzes structured requirements and design documents to find contradictions — for example, requirements that contradict each other or conditions that can never be jointly satisfied. We need to evaluate the LLM's ability to perform logical analysis over semi-structured technical prose.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	1	Serious concern	high
Input Content ■	1	Serious concern	high
Input Form ✓	3	Moderate gaps	high
Output Ontology ■	1	Serious concern	high
Output Content ✓	2	Significant gaps	high
Output Form ■	1	Serious concern	high
Average	1.5		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

FOLIO is structurally misaligned with the US Defense and High-Reliability Systems Engineering deployment on five of the six validity dimensions. The benchmark probes closed-world, fully-explicit, domain-neutral, single-context logical reasoning, with bare True/False/Unknown classification scored by accuracy. The deployment requires open-world, implicit-premise, domain-grounded, cross-document contradiction detection with natural-language explanations, confidence grades, and resolution paths for uncertain findings. Input Ontology, Input Content, Output Ontology, and Output Form score 1 (major violations); Output Content scores 2 (labels correct for the closed-world construct but reflect FOL-provability rather than engineering judgment, plus tangible annotation errors observable in sample); only Input Form scores 3, since both benchmark and deployment are text-only English even though length and register differ substantially. FOLIO accuracy

should be treated as a weak lower-bound proxy for general logical-reasoning capability — useful for filtering models that fail at multi-hop closed-world inference — but cannot serve as a predictive measure of deployment performance on any of the deployment's high-priority dimensions (IO, IC, OO, OF). The regulatory environment (CMMC, ITAR, CUI) adds an orthogonal deployment constraint that further restricts which models can be evaluated in compliant infrastructure.

Practical Guidance

What This Benchmark Measures

FOLIO measures a model's ability to perform closed-world, multi-hop deductive inference over fully-explicit narrative premises in English, classifying conclusions as True/False/Unknown by FOL semantics. It exercises broad logical-operator coverage (negation, conjunctions, disjunctions, implication, quantifiers, equality) and reasoning depths up to 7 hops. For the deployment, FOLIO can serve as a baseline filter on classical-logic competence — informative for the crisp-contradiction sub-task (explicit numeric conflicts), where Input Ontology's multi-hop closed-world chains are a useful lower-bound proxy.

Construct Depth

FOLIO probes its target construct (closed-world FOL reasoning) deeply: 1,430 inference-engine-verified examples, balanced True/False/Unknown distribution, 5+ hop reasoning in 28.7% of examples, broad AST diversity, and rigorous annotation QC. However, this depth does not extend to the deployment's high-priority constructs. Implicit-premise inference (IO-HIGH), latent domain contradictions (IC-HIGH), structured explanation + confidence (OO-HIGH), and cross-document reasoning (OF-HIGH) are not probed at any depth — they are absent from the benchmark's task taxonomy, content register, output schema, and evaluation apparatus. A model could achieve high FOLIO accuracy while failing completely on the deployment's primary use case.

What Else You Need

Substantial supplementation is required across IO, IC, OC, and OF. For implicit-premise document NLI (closest legal-domain analog): ContractNLI [\[WEB-10\]](#) and LawngNLI [\[WEB-11\]](#). For multi-document long-context reasoning (OF gap): MDBench [\[WEB-16\]](#), LongBench v2 [\[WEB-17\]](#), and PRELUDE [\[WEB-18\]](#). For explanation faithfulness (OO/OF gap): the FaithLM framework [\[WEB-19\]](#) and contradiction-explanation literature [\[WEB-21\]](#). None of these proxies covers MIL-STD/IEEE requirements-language reasoning [\[WEB-13, WEB-22\]](#) — that gap can only be filled by a purpose-built test suite of real or synthetic defense specification contradictions, reviewed by practicing systems engineers, ideally constructed to exercise latent thermal/timing/bandwidth contradiction patterns. Human-expert validation is methodologically necessary rather than optional, given that LLM explanation faithfulness measurement is itself an unresolved research problem [\[WEB-20, WEB-24\]](#).

ID	Evidence
WEB-10	ContractNLI is the closest structural analog for implicit-premise document NLI, but in the legal domain rather than engineering (https://www.emergentmind.com/topics/contractnli)
WEB-11	https://arxiv.org/abs/2212.03222
WEB-13	Space mission paper explicitly notes lack of standardized datasets for requirements verification — confirms no off-the-shelf engineer-annotated benchmark (https://arxiv.org/abs/2503.14130)
WEB-16	MDBench targets multi-document reasoning — usable as supplementary proxy for cross-document deployment capability (https://arxiv.org/abs/2506.14927)
WEB-17	LongBench v2 covers multi-document tasks with median 54K-word contexts — partial proxy for OF gap (https://arxiv.org/abs/2412.15204)
WEB-18	https://arxiv.org/abs/2508.09848
WEB-19	FaithLM evaluates natural-language explanation faithfulness — capability FOLIO does not assess (https://arxiv.org/abs/2402.04678)
WEB-20	All tested faithfulness metrics often fail to surpass a random baseline — explanation quality measurement is unresolved (https://arxiv.org/abs/2502.18848)

WEB-21	Reconciliatory contradiction explanation benchmark across 18 LLMs finds 'most models achieve limited success' (https://arxiv.org/abs/2603.22735)
WEB-22	NLP requirements inconsistency work (IEEE RE 2023) confirms a gap in standardized benchmarks for INCOSE-style requirements (https://medium.com/@dsobczynski88/applying-nlp-and-ai-to-improve-quality-of-software-requirement-statements-using-incose-guide-to-bd57ce2db14f)
WEB-24	ACL 2024 faithfulness research confirms LLM explanations frequently unfaithful to actual reasoning (https://www.lgresearch.ai/blog/view?seq=485)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's input ontology is structurally opposite to the deployment's core challenge. The benchmark is explicitly built around closed-world, fully-explicit premise sets where models 'decide the correctness of conclusions given a world defined by the premises' [\[Q15\]](#), and it explicitly excludes temporal and modal logics as 'beyond the scope' [\[Q139, Q140\]](#). The deployment's central task — recognizing latent conflicts where one or both relevant premises live implicitly in referenced MIL-STD/IEEE standards or shared engineering conventions — has no corresponding category in FOLIO's taxonomy. Dataset analysis confirms this without exception: every sampled example provides a complete self-contained premise set with no implicit-premise inference required [\[DATASET-D1, D3, D5\]](#). The deliberate exclusion of temporal/modal logic is particularly damaging since timing and conditional state requirements are first-class concerns in engineering specs. Cross-document/multi-context reasoning, the deployment's primary use case per Q4 elicitation, is also absent from FOLIO's task structure.

ID	Evidence
Q4	"Our experiments on FOLIO systematically evaluate the FOL reasoning ability of supervised fine-tuning on medium-sized language models." (p.1)
Q15	"We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises." (p.1)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)
Q140	"Consequently, they are beyond the scope of the definition of first-order logic used in our dataset." (p.13)
D1	No plants are fungi. Mushrooms are fungi. — Minimal 2-premise syllogism; fully explicit closed-world premises
D3	A project is written either in C++ or Python. If Sam does a project written in Python, he will not use a Mac. — Self-contained programming scenario; all relevant constraints explicitly stated in premises
D5	Everyone who is entitled to national social insurance coverage can have their medical bills partially covered. All PRC nationals are entitled to national social insurance coverage. — International policy scenario; all rules fully spelled out

Strengths

- Operator coverage is broad — negation, conjunction, inclusive/exclusive disjunction, implication, universal and existential quantifiers, and equality are all represented [\[Q138, DATASET-D12, D15\]](#), which provides at least minimal exposure to the logical structures (biconditionals, disjunctive requirements) that appear in specs.
- Reasoning depth diversity is substantive: 28.7% of examples require 5+ hops [\[Q60\]](#), and the sample confirms multi-hop chains of 4–6 universal conditionals [\[DATASET-D13, D6\]](#), providing a lower-bound proxy for explicit-chain reasoning capability.
- Includes both supervised fine-tuning and few-shot prompting evaluation regimes [\[Q82, Q88\]](#), plus advanced CoT/self-consistency/ToT and logic-specialized methods [\[Q90, Q91\]](#), which span the prompting strategies a deployment would likely use.

ID	Evidence
Q60	"As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1." (p.5)

Q82	"We test the logical reasoning capabilities of LMs using fully supervised fine-tuning and few-shot prompting." (p.6)
Q88	"For open-source models, we test LLaMA-13B and LLaMA-70B (Touvron et al., 2023), GPT-NeoX-20B (Black et al., 2022); for proprietary models we test GPT-3 (Brown et al., 2020), GPT-3.5-Turbo and GPT-4 (OpenAI et al., 2023) using prompts with 8 examples." (p.7)
Q90	"The prompting strategies include chain-of-thought (CoT) prompting (Wei et al., 2022b), chain-of-thought prompting with self-consistency (Wang et al., 2023) and tree-of-thought prompting (Yao et al., 2023)." (p.7)
Q91	"We also test recent methods specifically designed for logical reasoning: Logic-LM (2023), LINC (Olausson et al., 2023) and DetermLR(Sun et al., 2023), using GPT-4 as the base model." (p.7)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
D6	Everything in Size Town is big or small. All big things in Size Town are heavy. — Abstract taxonomy story; no real-world domain knowledge required
D12	$\forall x (\text{InThisTechCompany}(x) \wedge \text{Consistent}(x) \wedge \text{StickTo}(x, \text{theirRegularRoutine}) \rightarrow \neg \text{Like}(x, \text{surprise}))$ — FOL parallel annotation; shows explicit closed-world FOL encoding
D13	No trick-shot artist in Yale's varsity team struggles with half court shots. — Multi-premise sports scenario; label False from fully explicit premises
D15	If Clyde is not focused on futuristic and vocational subjects, then he is neither focused on futuristic and vocational subjects nor enjoys dressing up in old-fashioned and historical period clothing. — Complex conditional with exclusive disjunction; no domain knowledge required

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories per elicitation include: implicit-premise inference from referenced standards (Q1 elicitation), latent contradiction detection requiring domain reasoning (Q2 elicitation), cross-document conflict tracing (Q4 elicitation), and confidence-graded uncertainty with resolution paths (Q3 elicitation, Q6 gap). None of these are represented in FOLIO's two-task ontology of NL reasoning and NL-FOL translation [Q63].

IO-2: FOLIO omits every high-priority deployment category. The benchmark defines closed-world reasoning over self-contained premises [Q15] and explicitly excludes temporal and modal logics [Q139, Q140] — directly relevant to timing and conditional-state requirements. Dataset sample confirms no example involves implicit-premise inference [DATASET-D1, D3, D5]. Confirmed external gap: no benchmark targets MIL-STD/IEEE implicit-premise reasoning [WEB-13 — space mission requirement verification paper notes 'benchmarking is difficult due to the lack of standardized datasets'].

IO-3: The NL-FOL translation sub-task [Q67] and academic FOL-notation focus introduce categories not directly relevant to deployment, which targets natural-language explanation rather than FOL translation. Not fatal — accuracy on the NL reasoning sub-task can be reported in isolation — but consumes evaluation weight on a capability irrelevant to deployment value.

IO-4: Category gaps harming content validity: (a) no open-world / implicit-premise category, (b) no cross-document reasoning category, (c) no temporal/modal logic category, (d) no actionable-uncertainty category (Unknown means formal undecidability [Q66], not 'what would resolve this'), (e) no domain-specific latent-contradiction category. Combined, these gaps represent construct underrepresentation of the deployment's primary tasks.

ID	Evidence
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Q1	"FOLIO consists of 1,430 examples (unique conclusions), each paired with one of 487 sets of premises used to deductively reason for the validity of each conclusion." (p.1)
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q3	"In addition to the main NL reasoning task, NL-FOL pairs in FOLIO constitute a new NL-FOL translation dataset." (p.1)
Q4	"Our experiments on FOLIO systematically evaluate the FOL reasoning ability of supervised fine-tuning on medium-sized language models." (p.1)
Q6	"Our results show that a subset of FOLIO presents a challenge for one of the most capable Large Language Model (LLM) publicly available, GPT-4." (p.1)
Q15	"We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises." (p.1)
Q63	"We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation." (p.6)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q67	"We propose a new natural language to first-order logic translation (NL-FOL translation) task alongside our reasoning dataset." (p.6)
Q139	"Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics." (p.13)
Q140	"Consequently, they are beyond the scope of the definition of first-order logic used in our dataset." (p.13)
D1	No plants are fungi. Mushrooms are fungi. — Minimal 2-premise syllogism; fully explicit closed-world premises
D3	A project is written either in C++ or Python. If Sam does a project written in Python, he will not use a Mac. — Self-contained programming scenario; all relevant constraints explicitly stated in premises
D5	Everyone who is entitled to national social insurance coverage can have their medical bills partially covered. All PRC nationals are entitled to national social insurance coverage. — International policy scenario; all rules fully spelled out
WEB-10	ContractNLI is the closest structural analog for implicit-premise document NLI, but in the legal domain rather than engineering (https://www.emergentmind.com/topics/contractnli)
WEB-13	Space mission paper explicitly notes lack of standardized datasets for requirements verification — confirms no off-the-shelf engineer-annotated benchmark (https://arxiv.org/abs/2503.14130)

Information Gaps

- No correlation study between FOLIO accuracy and capability on implicit-premise or cross-document benchmarks (e.g., ContractNLI, MDBench).

Requires Expert Verification

- Whether FOLIO's multi-hop chain coverage is a useful lower-bound proxy for engineering crisp-contradiction detection requires domain-expert validation against representative specs.

Remediation

Gap 1: FOLIO has no task category for open-world / implicit-premise reasoning or cross-document tracing — directly opposite to the deployment's core challenge.

Recommendation: Construct a purpose-built test suite (~100–300 examples) of synthetic or de-classified defense spec excerpts where the relevant constraint lives in a referenced MIL-STD/IEEE/ICD section. Pair each example with a target finding written by a practicing systems engineer. Use FOLIO only as a baseline filter; do not treat FOLIO accuracy as evidence of deployment readiness.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's content is general-knowledge narrative prose drawn from Wikipedia seeds [\[Q34\]](#) and syllogism templates [\[Q42\]](#), with a vocabulary of 4,351 words [\[Q62\]](#) curated to avoid identity bias [\[Q47, Q127\]](#) and stay close to real-world facts [\[Q126\]](#). Dataset analysis confirms zero engineering-domain content across the sampled 57 examples: topics span sports, consumer products, biology, film, pop culture, and policy [\[DATASET-D9, D10, D22, D23, D24\]](#), with no MIL-STD/IEEE vocabulary, no tolerance/timing/thermal/bandwidth content, and no requirements-language modal verbs ('shall'/'should'). This produces systematic construct-irrelevant variance relative to the deployment's highest-priority targets — latent contradictions in thermal budgets, timing margins, and bus bandwidth [\[elicitation Q2\]](#). Web search confirms no available benchmark targets this content register [\[WEB-13, WEB-22\]](#). The WikiLogic contamination concern [\[Q104\]](#) further weakens the evidentiary value of FOLIO scores for novel technical content.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q34	"WikiLogic: annotation from scratch using Wikipedia articles as seeds. At this annotation stage, the annotators are asked to select random Wikipedia pages by repeatedly using the Wikipedia Special Random link. The Wikipedia articles are used to develop ideas for topics to write new stories." (p.4)
Q42	"In this approach, we first generate logically valid stories, which are templates containing abstract categories and entities, by combining multiple syllogisms into a single story template: the conclusion of one syllogism is used as a premise for the next syllogism." (p.4)
Q47	"Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion." (p.4)
Q62	"Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories." (p.5)
Q104	"Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data." (p.8)
Q126	"We rewrote those that are not reflective of well-established scientific, historical, or legal facts." (p.13)
Q127	"We took out stories that had strongly opinionated language and contained gender, racial, and classist biases." (p.13)
D9	Everyone that knows about breath-first-search knows how to use a queue. If someone is a seasoned software engineer interviewer at Google, then they know what breath-first-search is. — CS/software domain; all premises explicit and domain-neutral
D10	The world's only major large passenger aircraft manufacturers are Boeing and Airbus. All American Airlines planes are from the world's major large passenger aircraft manufacturers. — Aviation adjacent; but no technical specifications, tolerances, or implicit standards
D22	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination. — Contemporary brand reference; all premises fully stated
D23	Everyone at the business conference who is an ardent communist prefers state ownership of the means of production. — Political/ideological content (communist, planned economy); explicitly stated in premises
D24	The only types of mammals that lay eggs are either platypuses or echidnas. ... Grebes are not platypuses and also not echidnas. — Biology facts used as premises; verifiable real-world knowledge

WEB-13	Space mission paper explicitly notes lack of standardized datasets for requirements verification — confirms no off-the-shelf engineer-annotated benchmark (https://arxiv.org/abs/2503.14130)
WEB-22	NLP requirements inconsistency work (IEEE RE 2023) confirms a gap in standardized benchmarks for INCOSE-style requirements (https://medium.com/@dsobczynski88/applying-nlp-and-ai-to-improve-quality-of-software-requirement-statements-using-incose-guide-to-bd57ce2db14f)

Strengths

- Content is intentionally screened against identity bias [Q47, Q127], reducing one class of construct-irrelevant variance that could plausibly affect engineering-domain deployment.
- Logical complexity and AST diversity are genuine — 4,351-word vocabulary, 28.7% of examples requiring 5+ hops [Q60, Q62] — providing a substantively non-trivial logical-reasoning probe even if domain-irrelevant.
- Commonsense premises required for FOL reasoning are explicitly added as additional premises [Q55], demonstrating awareness of implicit-knowledge issues — though this is solved by elimination (making everything explicit) rather than by testing implicit-knowledge invocation, which is exactly what the deployment needs.

ID	Evidence
Q47	“Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion.” (p.4)
Q55	“Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage.” (p.5)
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)
Q62	“Table 3 shows that our dataset has a vocabulary of 4,351 words, and the examples based on Wikipedia account for 74% of the total vocabulary even though the WikiLogic stories take up only 63% of the total number of stories.” (p.5)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Deployment requires inputs in normalized requirements-language register with modal verbs ('shall', 'should'), standards identifiers (MIL-STD-461, IEEE 15288), tolerance expressions, and interface parameter conventions. FOLIO contains none of this register — every sample example is general-purpose narrative English [DATASET-D9, D10, D22].

IC-2: Cultural alignment is partial — content is English-language and US-acceptable [DATASET-D11, D22, D26], but 'cultural fit' to the engineering professional culture is essentially zero: no examples involve the standards-driven, traceability-focused register engineers work in.

IC-3: Inputs require general Western/Anglophone cultural knowledge (TikTok, Meta, Yale, Hong Kong references — [DATASET-D11, D22, D26]). This is compatible with US deployment at the surface level, but irrelevant to engineering content. WikiLogic contamination [Q104] is a confound: LLMs may answer from pretraining rather than logical inference.

IC-4: INSUFFICIENT DOCUMENTATION — no evidence of recruiting systems engineers as annotators or reviewers; benchmark used CS/NLP/FOL experts [Q29, Q124, Q125], not aerospace/defense practitioners. Regional expert recruitment would be needed.

IC-5: Content issues harming content validity: (a) zero engineering-domain examples [DATASET-D9, D10, D22], (b) no normalized requirements-language register, (c) no MIL-STD/IEEE/INCOSE vocabulary, (d) Wikipedia contamination confound [Q104, DATASET-D26], (e) HybLogic template learnability [Q105] inflates measured performance on patterns absent from real specs.

ID	Evidence
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q104	"Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data." (p.8)
Q105	"However, fine-tuning RoBERTa-large achieves higher performance on HybLogic than on WikiLogic. This is likely because HybLogic is created from templates and some of the logical patterns can be learned during fine-tuning." (p.8)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)
D9	Everyone that knows about breath-first-search knows how to use a queue. If someone is a seasoned software engineer interviewer at Google, then they know what breath-first-search is. — CS/software domain; all premises explicit and domain-neutral
D10	The world's only major large passenger aircraft manufacturers are Boeing and Airbus. All American Airlines planes are from the world's major large passenger aircraft manufacturers. — Aviation adjacent; but no technical specifications, tolerances, or implicit standards
D11	TikTok is a social media application, and it is not ideal for preteens. — Consumer tech scenario; ¬(chat ⊕ computer_program) conclusion structure
D22	Everyone working at Meta has a high income. A person with a high income will not take a bus to their destination. — Contemporary brand reference; all premises fully stated
D26	Andy Chang is from Hong Kong. — Specific named person (likely real individual); Wikipedia-seeded
WEB-13	Space mission paper explicitly notes lack of standardized datasets for requirements verification — confirms no off-the-shelf engineer-annotated benchmark (https://arxiv.org/abs/2503.14130)
WEB-22	NLP requirements inconsistency work (IEEE RE 2023) confirms a gap in standardized benchmarks for INCOSE-style requirements (https://medium.com/@dsobczynski88/applying-nlp-and-ai-to-improve-quality-of-software-requirement-statements-using-incose-guide-to-bd57ce2db14f)

Information Gaps

- Whether the test split (226 examples, not publicly visible) contains any engineering-domain content — sample was train-only.
- Whether full annotator pool includes anyone with systems-engineering background — documentation suggests no [Q108, Q122].

Requires Expert Verification

- Whether any FOLIO content patterns transfer at all to MIL-STD/IEEE requirements-language reasoning would require empirical testing with practicing systems engineers.

Remediation

Gap 1: Zero engineering-domain vocabulary or 'shall/should' requirements-language register in FOLIO; systematic construct-irrelevant variance for the deployment.

Recommendation: Augment evaluation with a small, curated corpus of requirements-language examples drawn from publicly available NASA, INCOSE, or unclassified DoD specs, annotated for crisp and latent contradictions by systems engineers. Even 50–100 high-quality engineer-validated examples will provide more signal than FOLIO's 1,430

domain-neutral stories for the deployment-specific construct.

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is the one dimension where FOLIO partially aligns with the deployment. Both are text-only English [Q22, DATASET-D4, D8], with no modality or script mismatch. FOLIO's surface conventions are carefully standardized: 'either-or' for exclusive disjunction [Q130], 'A or B, or both' for inclusive [Q131], grammar-checked with Grammarly [Q49], reviewed for naturalness [Q50], and ambiguity actively eliminated [Q51]. However, two substantial form mismatches remain. First, FOLIO inputs are short self-contained narratives (sampled premises range from 2 to ~9 sentences, ~50–200 tokens [DATASET-D1, D2]), while deployment documents are multi-thousand-word, multi-section specs (20,000–250,000+ words; 27K–100K+ tokens [WEB-6, WEB-7]). Second, FOLIO's deliberate elimination of ambiguity [Q51] produces inputs cleaner than real specs, which contain modal verbs, cross-references, and tolerance expressions absent from FOLIO. Recent long-context research [WEB-7] shows accuracy degradation from ~95% to ~60% at long contexts — a regime FOLIO cannot probe. The form mismatch is real but not categorical, so the dimension scores higher than IO/IC/OO/OF.

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q49	"Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly." (p.4)
Q50	"Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness." (p.4)
Q51	"We also eliminate natural language ambiguity when it is possible." (p.4)
Q130	"We always use "either-or" to express exclusive disjunction." (p.13)
Q131	"We use either "A or B" or "A or B, or both" to express inclusive disjunction." (p.13)
D1	No plants are fungi. Mushrooms are fungi. — Minimal 2-premise syllogism; fully explicit closed-world premises
D2	People in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. — Multi-premise narrative about a tech company employee; all predicates fully stated
D4	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. A country cannot simultaneously regulate the exchange rate and successfully control inflation. — Policy/economics domain; fully explicit premises; no external standard referenced
D8	Everything is either outside the solar system or in the solar system. Nothing outside the solar system has the Sun as its star. — Scientific premises fully stated; no reference to external standards
WEB-6	250-page technical spec can exceed 60,000 tokens; ~20,000-word SRD+ICD pair yields ~27,000 tokens (https://www.devtoolkit.cloud/blog/llm-context-windows-explained-why-size-matters)
WEB-7	Long-context accuracy degrades from ~95% to ~60% on frontier models as input length grows (https://blog.bytebytego.com/p/a-guide-to-context-engineering-for)

Strengths

- Text-only English with no modality mismatch [Q22, DATASET-D4, D8] — directly aligns with deployment's text-only English interface.

- Russell & Norvig FOL syntax standard [Q52] is widely taught in CS curricula and would be familiar to engineering reviewers with formal training.
- Premise ordering does not encode label information [Q158, Q159, Q160] — accuracy changes ~1% under shuffling — so the benchmark is robust to one class of input-form artifact.
- Story-level train/val/test split [Q78, Q79] prevents premise leakage, supporting external validity for the form.

ID	Evidence
Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q52	"We adopt the FOL definitions and syntax most widely used in the AI community (Russell and Norvig, 2010)." (p.5)
Q78	"We split FOLIO by 70%/15%/15% split for the train/validation/test sets with 1,001/203/226 examples respectively." (p.6)
Q79	"We split by story so that models are evaluated on unseen stories." (p.6)
Q158	"To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models." (p.15)
Q159	"We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises." (p.15)
Q160	"This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions." (p.15)
D4	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency. A country cannot simultaneously regulate the exchange rate and successfully control inflation. — Policy/economics domain; fully explicit premises; no external standard referenced
D8	Everything is either outside the solar system or in the solar system. Nothing outside the solar system has the Sun as its star. — Scientific premises fully stated; no reference to external standards

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions partially match: both are text-only English, but FOLIO inputs are short stories (~50–200 tokens) [DATASET-D1, D2] while deployment requires multi-thousand-word, multi-section documents (27K–100K+ tokens per [WEB-6]). The benchmark does not exercise long-context capability at all.

IF-2: Regional infrastructure supports text input trivially. No capture-specification mismatch. However, on-premises/FedRAMP High deployment constraints driven by CUI/ITAR [WEB-3, WEB-4, WEB-9] mean models with very large context windows may not be available in compliant deployment configurations.

IF-3: Domain-specific form differences: (a) FOLIO eliminates NL ambiguity [Q51], deployment documents are full of modal verbs and conditional language; (b) FOLIO uses standardized either-or/inclusive-or conventions [Q130, Q131], deployment uses normalized 'shall/should' register; (c) FOLIO has parallel FOL annotations [Q65], deployment has no FOL parallel — engineers reason directly on NL.

IF-4: Form mismatches harming external validity: (a) input length 2–3 orders of magnitude shorter than deployment [DATASET-D1, D2, WEB-6]; (b) ambiguity actively eliminated [Q51] whereas deployment must handle ambiguity; (c) no multi-document structural form; (d) no normalized requirements-language register.

ID	Evidence
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Q22	"We use formal logic, i.e., FOL to ensure the logical validity of the examples written in NL and propose a new NL-FOL translation task." (p.2)
Q51	"We also eliminate natural language ambiguity when it is possible." (p.4)
Q65	"All NL stories are annotated with parallel FOL stories SF, which are sets of FOL formulas consisting of n premises $PF = \{pf1, pf2, \dots, pfn\}$ and m conclusions $HF = \{hf1, hf2, \dots, hfm\}$." (p.6)
Q130	"We always use "either-or" to express exclusive disjunction." (p.13)
Q131	"We use either "A or B" or "A or B, or both" to express inclusive disjunction." (p.13)
D1	No plants are fungi. Mushrooms are fungi. — Minimal 2-premise syllogism; fully explicit closed-world premises
D2	People in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. — Multi-premise narrative about a tech company employee; all predicates fully stated
WEB-3	https://www.dwt.com/blogs/privacy--security-law-blog/2025/09/defense-department-cybersecurity-cmmc-final-rule
WEB-4	https://www.kiteworks.com/regulatory-compliance/itar-ai-agents-compliance-gap/
WEB-6	250-page technical spec can exceed 60,000 tokens; ~20,000-word SRD+ICD pair yields ~27,000 tokens (https://www.devtoolkit.cloud/blog/llm-context-windows-explained-why-size-matters)
WEB-7	Long-context accuracy degrades from ~95% to ~60% on frontier models as input length grows (https://blog.bytebytego.com/p/a-guide-to-context-engineering-for)
WEB-9	https://www.kiteworks.com/cmmc-compliance/military-technology-contractors/

Information Gaps

- No data on FOLIO performance under controlled length perturbations (e.g., padding stories with distractor premises) — would be informative for length sensitivity.

Requires Expert Verification

- Whether the elimination of ambiguity [Q51] makes FOLIO performance an upper bound on real-spec performance — domain experts could validate by reviewing model outputs on real specs.

Remediation

Gap 1: FOLIO inputs are short ambiguity-free narratives (50–200 tokens); deployment inputs are 27K–100K+ token multi-section specifications where LLM reasoning quality is known to degrade.

Recommendation: Include explicit long-context stress tests in the deployment evaluation: replicate or extend the Chroma 2025-style length-degradation analysis [WEB-7] with representative spec documents to characterize the operating context-length envelope. Pair this with infrastructure planning (FedRAMP High / on-premises) per the CMMC/ITAR constraints [WEB-1, WEB-4].

ID	Evidence
WEB-1	https://www.federalregister.gov/documents/2024/10/15/2024-22905/cybersecurity-maturity-model-certification-cmmc-program
WEB-4	https://www.kiteworks.com/regulatory-compliance/itar-ai-agents-compliance-gap/
WEB-7	Long-context accuracy degrades from ~95% to ~60% on frontier models as input length grows (https://blog.bytebytego.com/p/a-guide-to-context-engineering-for)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output ontology — True/False/Unknown classification labels [Q36, Q66, Q94] — is fundamentally misaligned with deployment requirements. The deployment explicitly demands natural-language explanations of the conflict mechanism paired with a confidence grade [Q3 elicitation, output_requirements.required_output_form], with bare classification 'explicitly non-actionable' for engineers reviewing hundreds of requirements. Dataset analysis confirms every example has exactly one of three string labels with no explanation, rationale, or confidence field [DATASET-D11, D25]. Furthermore, FOLIO's 'Unknown' label denotes formal undecidability from stated premises [Q66, DATASET-D25] — a fundamentally different concept from the deployment's notion of actionable uncertainty, which requires a statement of what additional information (referenced standard content, interface parameter) would resolve the ambiguity. An unexplained 'Uncertain' is treated as a tool failure in this deployment [output_requirements.uncertain_handling]. The output taxonomy violates both structural validity (the construct's structure is misrepresented — actionable contradiction explanation is reduced to a 3-way classification) and content validity (missing categories: explanation quality, calibrated confidence, resolution paths).

ID	Evidence
Q3	"In addition to the main NL reasoning task, NL-FOL pairs in FOLIO constitute a new NL-FOL translation dataset." (p.1)
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
D11	TikTok is a social media application, and it is not ideal for preteens. — Consumer tech scenario; ¬(chat ⊕ computer_program) conclusion structure
D25	Each building is tall. Everything tall has height. — Minimal 2-premise story; "All buildings are magnificent" — Uncertain since 'magnificent' not derivable

Strengths

- Three-way label structure (not binary) [Q36, DATASET-D14, D25] is preferable to a pure binary, since it minimally distinguishes 'definitively conflicts' from 'undetermined' — a partial echo of the deployment's triage need.
- Labels are grounded in FOL semantics verified by an inference engine [Q2, Q57], so label correctness for the benchmark's intended construct is high.
- FOL operator coverage is comprehensive [Q138] — negation, conjunctions, disjunctions, implication, quantifiers, equality — supporting unambiguous derivation of truth values [Q136, Q137].

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q36	"Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example)." (p.4)

Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q136	"FOL, as a logical form, is a more explicit logical representation than its NL counterpart and can be used as input to an FOL prover in order to obtain the exact truth values for the conclusions." (p.13)
Q137	"FOL has no ambiguity while ambiguity can occur at various levels of NLP." (p.13)
Q138	"We include the following operators: negation $\neg$, conjunction $\wedge$, disjunction $\vee$, implication $\rightarrow$, universal quantifier $\forall$, existential quantifier $\exists$, equal $=$." (p.13)
D14	All Olympic gold medal winners are professional athletes. No full-time scientists spend the majority of their time on sports. — 6-premise chain with compound conditional; inference-engine verified True
D25	Each building is tall. Everything tall has height. — Minimal 2-premise story; "All buildings are magnificent" — Uncertain since 'magnificent' not derivable

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The True/False/Unknown taxonomy [Q66] is regionally relevant only in the most abstract sense (the labels themselves are not culturally biased). But for the deployment, the relevant 'output category' is structured natural-language explanation + confidence grade — a category entirely absent from FOLIO.

OO-2: Missing categories specific to deployment: (a) natural-language explanation of conflict mechanism, (b) confidence grade for triage [output_requirements.triage_function], (c) resolution-path statement for uncertain findings, (d) traceability citation (REQ-IDs, document/section references) per cultural_and_professional_norms_notes. All four are essential to the deployment and none is in FOLIO's output ontology [DATASET-D11, D25].

OO-3: The 'Unknown' label encodes a non-deployment assumption: closed-world formal undecidability [Q66, DATASET-D25]. This is the opposite of the deployment's actionable-uncertainty notion. Models optimized on FOLIO learn to recognize closed-world undecidability, not to articulate what missing information would resolve an open-world conflict [output_requirements.uncertain_handling].

OO-4: Stakeholder-driven taxonomy redesign is necessary. The deployment requires a fundamentally different output schema (structured explanation + confidence + resolution path + traceability). FOLIO cannot be adapted by relabeling — the entire output structure must be replaced.

OO-5: Taxonomy issues harming validity: (a) bare classification labels reduce a generative, explanatory task to a 3-way categorization [DATASET-D11]; (b) Unknown $\neq$ deployment's Uncertain [Q66, DATASET-D25]; (c) no confidence/calibration category [WEB-20 confirms calibration is a broader unresolved issue]; (d) no mechanism to assess explanation faithfulness [WEB-19, WEB-20, WEB-24].

ID	Evidence
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
D11	TikTok is a social media application, and it is not ideal for preteens. — Consumer tech scenario; $\neg(\text{chat} \oplus \text{computer_program})$ conclusion structure
D25	Each building is tall. Everything tall has height. — Minimal 2-premise story; "All buildings are magnificent" — Uncertain since 'magnificent' not derivable

WEB-19	FaithLM evaluates natural-language explanation faithfulness — capability FOLIO does not assess (https://arxiv.org/abs/2402.04678)
WEB-20	All tested faithfulness metrics often fail to surpass a random baseline — explanation quality measurement is unresolved (https://arxiv.org/abs/2502.18848)
WEB-24	ACL 2024 faithfulness research confirms LLM explanations frequently unfaithful to actual reasoning (https://www.lgresearch.ai/blog/view?seq=485)

Information Gaps

- Whether composite metrics combining FOLIO accuracy with separately-measured explanation-quality scores correlate with deployment-task performance — no such study exists.

Requires Expert Verification

- Definition of acceptable confidence-grade calibration in the deployment context (e.g., what 'high confidence' threshold maps to engineering action) requires program-office input.

Remediation

Gap 1: Bare True/False/Unknown classification cannot capture the deployment's required output (NL explanation + confidence grade + resolution path).

Recommendation: Redesign the output schema for any deployment evaluation: require structured outputs with (a) NL conflict-mechanism explanation citing specific requirement IDs, (b) calibrated confidence grade for triage, and (c) for uncertain findings, an explicit statement of what missing information would resolve the ambiguity. Evaluate explanation quality using a faithfulness-aware protocol (e.g., FaithLM-style [WEB-19]) plus blind engineer review.

ID	Evidence
WEB-19	FaithLM evaluates natural-language explanation faithfulness — capability FOLIO does not assess (https://arxiv.org/abs/2402.04678)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's ground-truth labels are derived from automated FOL inference engine verification [Q2, Q57] over annotations written by CS students/researchers with formal FOL training [Q31, Q108, Q122, Q123]. The human expert baseline (95.98% accuracy) versus non-expert (61.82%) [Q111] confirms labels reflect what is logically provable from stated premises by FOL-trained humans. For the deployment, this introduces two convergent-validity concerns. First, annotators have no documented systems-engineering, aerospace, or defense expertise [Q108, Q122] — the ground truth reflects logical provability, not whether a practicing engineer would recognize a latent domain-convention conflict. Second, dataset analysis surfaces tangible annotation quality issues in the small sample: name mismatches between NL and FOL [DATASET-D18], an NL/FOL semantic contradiction [DATASET-D17], and several FOL parenthesis errors [DATASET-D16, D27, D28]. With ~5 identifiable issues in a 57-example sample, the error rate may be non-trivial, weakening the evidentiary value of FOLIO accuracy scores as a deployment proxy. The dimension scores 2 (not 1) because labels are correct in the closed-world FOL sense — they are wrong as a proxy for engineering judgment, not internally inconsistent in their own taxonomy.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q31	"All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria." (p.4)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q108	"Our expert annotators are computer science college students familiar with FOL." (p.9)
Q111	"Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%." (p.9)
Q122	"Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.12)
Q123	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing." (p.12)
D16	Customer(lily) $\wedge$ In(lily, jamesFamily $\wedge$ WatchIn(lily, tv, cinema) — Parenthesis error in FOL annotation for Lily's premise
D17	Matt does not invest in the public stock market regularly. Matt likes financial risks. — Contradictory premise pair (Ex 57 NL says Matt does NOT invest but FOL says InvestInRegularly(matt, publicStockMarket))
D18	Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises. ... $\neg(\text{ProfessionalBasketballPlayer}(\text{jim}) \oplus \text{NeverExercises}(\text{jim}))$ — NL says "John," FOL says "jim" — name inconsistency between NL and FOL
D27	$\forall x (\text{SpendAt}(x, \text{aLotOfMoney, casino}) \vee (\exists y (\neg(y=\text{casino}) \wedge \text{BettingGame}(y) \wedge \text{SpendAt}(x, \text{aLotOfMoney, y})) \rightarrow \text{AtRiskOf}(x, \text{gamblingAddiction}))$ — FOL missing closing parenthesis — syntactic issue in annotation
D28	(Knows(dan, dune) $\wedge$ ScienceFiction(dune)) $\vee$ ProvedToBe(dune, false)) — Extra closing parenthesis in FOL — syntactic issue

Strengths

- Inference-engine verification of label correctness [Q2, Q57] provides strong intra-construct ground truth — labels are not subjective and have been programmatically checked.

- Annotator selection criteria are explicit and rigorous: native or near-native English speakers with formal FOL coursework [Q26, Q27, Q28], multi-stage NL and FOL quality control by NLP/CL and FOL experts respectively [Q29, Q124, Q125].
- Substantial annotation investment: 980 man-hours across six stages [Q32], with 39.2% of stories rewritten after review [Q48] — suggests strong internal quality discipline.
- Three-way label distribution is reasonably balanced [Q94, DATASET-D14, D25] — models cannot trivially collapse to a binary classifier.

ID	Evidence
Q2	"The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine." (p.1)
Q26	"Given the complexities of such annotations, we selected annotators based on a few important criteria to ensures that our dataset is annotated with the highest level of precision and expertise, reflecting the complexity and nuance required for first-order logical reasoning." (p.3)
Q27	"Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.3)
Q28	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-" (p.3)
Q29	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.4)
Q32	"We develop our dataset in six stages: WikiLogic collection, HybLogic collection, NL quality control, FOL quality control, NL-FOL alignment and FOL verification, spending 980 man-hours in total." (p.4)
Q48	"Toward these objectives, we manually screened all stories and found that 39.2% of the stories suffer from at least one of these issues. We implemented a detailed protocol to rewrite these stories." (p.4)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q94	"The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively." (p.7)
Q124	"At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved." (p.12)
Q125	"For the FOL quality check, only annotators who are experts in first-order logic are involved." (p.12)
D14	All Olympic gold medal winners are professional athletes. No full-time scientists spend the majority of their time on sports. — 6-premise chain with compound conditional; inference-engine verified True
D25	Each building is tall. Everything tall has height. — Minimal 2-premise story; "All buildings are magnificent" — Uncertain since 'magnificent' not derivable

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Labels reflect FOL-provability judgments by CS/FOL-expert annotators [Q108, Q122], not systems-engineering stakeholder perspectives. For deployment-relevant 'latent contradictions involving implicit domain constraints,' the labels do

not encode the practicing engineer's judgment.

OC-2: Likely substantial disagreement: a systems engineer reviewing a real spec would invoke MIL-STD/IEEE conventions and physical constraints not in the premise set. FOLIO annotators were instructed to add necessary commonsense as explicit premises [Q55] — solving the problem by elimination, not by testing implicit-knowledge invocation. Confirmed structural gap [WEB-13].

OC-3: Documentation discloses annotator demographics: CS undergraduate/graduate students and researchers [Q16, Q31], native or near-native English [Q27], FOL coursework [Q28, Q123]. No aerospace/defense/ICD background documented. Datasheet-style disclosure is partial.

OC-4: Re-annotation by representative regional pool (US-based practicing systems engineers) would be necessary to assess deployment validity but is impractical at FOLIO scale; the more economical path is a purpose-built test suite of real or synthetic defense spec contradictions reviewed by domain experts.

OC-5: INSUFFICIENT DOCUMENTATION on aggregation. Paper does not describe inter-annotator agreement statistics by sub-population or whether minority-perspective disagreements were preserved; the FOL-verification step [Q57] arguably erases human disagreement entirely in favor of automated provability.

OC-6: Label issues harming convergent/external validity: (a) annotator pool lacks deployment-domain expertise [Q108, Q122]; (b) labels reflect closed-world provability, not open-world engineering judgment [Q66]; (c) observable annotation errors in sample (NL/FOL mismatches, FOL parenthesis issues) [DATASET-D17, D18, D16, D27]; (d) Wikipedia contamination [Q104] confounds label-quality interpretation for LLM evaluation.

ID	Evidence
Q16	"FOLIO is a high-quality and manually curated dataset, written by CS undergraduate and graduate students and researchers in academia and industry." (p.2)
Q27	"Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.3)
Q28	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-" (p.3)
Q31	"All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria." (p.4)
Q55	"Apart from checking whether NL and FOL express equivalent meanings, we also add necessary commonsense knowledge in both the NL and FOL premises. Sometimes humans do not write certain commonsense knowledge in the premises that is required in the FOL reasoning process, which is based solely on the premises given. We add such knowledge as additional premises at this stage." (p.5)
Q57	"Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine." (p.5)
Q66	"Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning." (p.6)
Q104	"Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data." (p.8)
Q108	"Our expert annotators are computer science college students familiar with FOL." (p.9)
Q122	"Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English." (p.12)
Q123	"They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing." (p.12)
D16	Customer(lily) $\wedge$ In(lily, jamesFamily $\wedge$ WatchIn(lily, tv, cinema) — Parenthesis error in FOL annotation for Lily's premise
D17	Matt does not invest in the public stock market regularly. Matt likes financial risks. — Contradictory premise pair (Ex 57 NL says Matt does NOT invest but FOL says InvestInRegularly(matt, publicStockMarket))
D18	Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises. ... $\neg(\text{ProfessionalBasketballPlayer}(\text{jim}) \oplus \text{NeverExercises}(\text{jim}))$ — NL says "John," FOL says "jim" — name inconsistency between NL and FOL
D27	$\forall x (\text{SpendAt}(x, \text{alotOfMoney}, \text{casino}) \vee (\exists y (\neg(y=\text{casino}) \wedge \text{BettingGame}(y) \wedge \text{SpendAt}(x, \text{aLotOfMoney}, y)) \rightarrow \text{AtRiskOf}(x, \text{gamblingAddiction})))$ — FOL missing closing parenthesis — syntactic issue in annotation

WEB-13

Space mission paper explicitly notes lack of standardized datasets for requirements verification — confirms no off-the-shelf engineer-annotated benchmark (<https://arxiv.org/abs/2503.14130>)

Information Gaps

- Annotation error rate on full dataset (sample suggests ~5-9% of examples have issues; test split not inspected).
- Inter-annotator agreement statistics are not reported in the documentation.

Requires Expert Verification

- Whether a defense systems engineer would assign the same labels to FOLIO examples as the CS-trained annotators — informative for understanding the label-construct gap.

Remediation

Gap 1: Annotators are CS/FOL experts, not systems engineers; ground-truth labels reflect closed-world provability rather than engineering domain judgment. Sample inspection also reveals scattered NL/FOL inconsistencies.

Recommendation: For the deployment's purpose-built test suite, recruit ≥ 3 practicing systems engineers per example for label consensus, retain dissent for ambiguous cases rather than aggregating it away, and explicitly capture whether the contradiction requires invocation of an implicit referenced-standard constraint. Treat FOLIO ground-truth as informative only for the closed-world sub-construct.

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: FOLIO | Context: US Defense and High-Reliability Systems Engineers

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's output form — accuracy on True/False/Unknown classification [Q80] plus SynV and ExcAcc for translation [Q73–Q76] — is fundamentally mismatched with deployment needs. The deployment requires structured natural-language explanations citing specific requirement IDs (e.g., 'REQ-SYS-0042 in the SRD conflicts with ICD-PWR-007 Section 3.2.1' per cultural_and_professional_norms_notes), a confidence grade for triage [output_requirements.triage_function], and resolution paths for uncertain findings — none of which FOLIO measures. Critically, FOLIO evaluates single isolated (premise_set, conclusion) pairs [DATASET-D21, D2], whereas the deployment's central use case is cross-document reasoning across multiple long-form documents simultaneously [Q4 elicitation, cross_document_reasoning_note]. There is no benchmark mechanism that probes multi-document context handling or traceability output. Auxiliary findings amplify the concern: GPT-4 is 31.82% below expert human accuracy [Q113], models systematically overpredict True [Q155, Q157] with False/Unknown accuracy only 54–62% [Q156], and performance degrades sharply beyond 3 reasoning hops [Q99] — all indicating that even the closed-world classification baseline is challenged, while the deployment requires capabilities entirely outside the benchmark's measurement apparatus. The authors themselves defer 'a more reliable metric' for translation to future work [Q77].

ID	Evidence
Q4	"Our experiments on FOLIO systematically evaluate the FOL reasoning ability of supervised fine-tuning on medium-sized language models." (p.1)
Q73	"Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV)." (p.6)
Q74	"The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2)." (p.6)
Q75	"Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion." (p.6)
Q76	"We define the accuracy of the output labels as the execution accuracy." (p.6)
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
Q99	"With few-shot prompting, GPT-3.5 and GPT-4 both perform much better on examples with a 0 ~ 3 reasoning depth, indicating that examples with a 4 ~ 7 reasoning depth pose a challenge to the SoTA LMs." (p.8)
Q113	"The expert and GPT-4 gap is 31.82%, suggesting significant room for model improvement." (p.9)
Q155	"Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown." (p.15)
Q156	"The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting." (p.15)
Q157	"They also tend to make more predictions of True than the other labels." (p.15)
D2	People in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. — Multi-premise narrative about a tech company employee; all predicates fully stated
D21	All Yale students at the event are college students. Everyone at the event is a Yale student or a Harvard student. — Classic syllogistic structure, fully explicit

Strengths

- Stability protocol — five randomly-sampled training sets with averaged accuracy [Q93] — produces robust point estimates rather than noisy single runs.
- Premise-shuffling experiment confirms minimal label-order artifacts (~1% accuracy change) [Q158, Q159, Q160], so accuracy reflects reasoning rather than positional heuristics.
- Story-level train/val/test split [Q78, Q79] prevents premise leakage, supporting external validity of the accuracy estimate within the benchmark's construct.
- Reporting breakdowns by reasoning depth [Q99] and by sub-pipeline (WikiLogic vs. HybLogic) [Q102, Q105] provides diagnostic granularity useful for capability profiling.

ID	Evidence
Q78	"We split FOLIO by 70%/15%/15% split for the train/validation/test sets with 1,001/203/226 examples respectively." (p.6)
Q79	"We split by story so that models are evaluated on unseen stories." (p.6)
Q93	"We run experiments on five randomly sampled sets of examples from the training set and report the average accuracy." (p.7)
Q99	"With few-shot prompting, GPT-3.5 and GPT-4 both perform much better on examples with a 0 ~ 3 reasoning depth, indicating that examples with a 4 ~ 7 reasoning depth pose a challenge to the SoTA LMs." (p.8)
Q102	"As shown in Table 6, in logical reasoning, GPT-3.5 and GPT-4 achieve substantially lower results on HybLogic than on WikiLogic and the result is slightly higher than chance." (p.8)
Q105	"However, fine-tuning RoBERTa-large achieves higher performance on HybLogic than on WikiLogic. This is likely because HybLogic is created from templates and some of the logical patterns can be learned during fine-tuning." (p.8)
Q158	"To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models." (p.15)
Q159	"We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises." (p.15)
Q160	"This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality mismatch is total. Deployment requires structured natural-language explanations + confidence grade + traceability citations [output_requirements]; FOLIO outputs a single label per example [Q80, DATASET-D11]. Single-context vs. multi-document mismatch [DATASET-D21, D2] is equally categorical.

OF-2: Not applicable — deployment is text-only [infrastructure_and_deployment_notes.interface_modality]; no speech output requirement.

OF-3: Not applicable — target users are professional engineers with high literacy in normalized requirements register; accessibility constraints are not the binding constraint here. The relevant accessibility analog is whether outputs are actionable for expert review, which they are not [DATASET-D11, D25].

OF-4: Form mismatches harming external validity: (a) classification accuracy is the only metric [Q80], cannot assess explanation quality [WEB-19, WEB-20]; (b) no multi-document evaluation [DATASET-D21]; (c) no confidence-calibration metric; (d) no traceability/citation metric; (e) NL-FOL translation metrics SynV and ExcAcc are acknowledged as imperfect [Q77] and target a capability deployment does not need.

ID	Evidence
Q77	"We leave for future work the design of a more reliable metric of NL-FOL translation." (p.6)
Q80	"We use accuracy for evaluating logical reasoning performance." (p.6)
D2	People in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. — Multi-premise narrative about a tech company employee; all predicates fully stated
D11	TikTok is a social media application, and it is not ideal for preteens. — Consumer tech scenario; $\neg(\text{chat} \oplus \text{computer_program})$ conclusion structure
D21	All Yale students at the event are college students. Everyone at the event is a Yale student or a Harvard student. — Classic syllogistic structure, fully explicit
D25	Each building is tall. Everything tall has height. — Minimal 2-premise story; "All buildings are magnificent" — Uncertain since 'magnificent' not derivable
WEB-19	FaithLM evaluates natural-language explanation faithfulness — capability FOLIO does not assess (https://arxiv.org/abs/2402.04678)
WEB-20	All tested faithfulness metrics often fail to surpass a random baseline — explanation quality measurement is unresolved (https://arxiv.org/abs/2502.18848)

Information Gaps

- Whether FOLIO accuracy correlates at all with explanation-quality or multi-document-reasoning capability — no such correlation study exists [gap_id 4].

Requires Expert Verification

- Whether MDBench or LongBench v2 scores serve as adequate supplementary proxies for the deployment's cross-document requirement requires domain validation.

Remediation

Gap 1: FOLIO measures single-example classification accuracy with no mechanism for multi-document or long-context evaluation, no confidence-calibration metric, and no traceability-citation metric.

Recommendation: Add MDBench [WEB-16] and LongBench v2 [WEB-17] as supplementary cross-document/long-context proxies. Build a multi-document evaluation harness using SRD+ICD pairs (real or synthetic) where the contradiction spans documents; report not only accuracy but explanation quality, confidence calibration (e.g., ECE), and citation accuracy (correct REQ-IDs cited).

ID	Evidence
WEB-16	MDBench targets multi-document reasoning — usable as supplementary proxy for cross-document deployment capability (https://arxiv.org/abs/2506.14927)
WEB-17	LongBench v2 covers multi-document tasks with median 54K-word contexts — partial proxy for OF gap (https://arxiv.org/abs/2412.15204)